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Where Minds Pollinate

Class 9 Biology

Chapter 10

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Unit no. 10

Topic: Evolution

1. What is variation? Explain the sources of variation.

Variation:

The term variation refers to the differences in traits displayed by members of the same species. For instance, although all domestic cats are members of the same species, they differ greatly in terms of size, coat colour, eye colour, hair length and other characteristics.

Importance of variation:

- Increasing their chances of survival
- Desirable variations help them be more advanced & immune
- Lead to genetic defect

Main sources of Variations:

- Genetic recombination results from crossing over during gamete development.
- Different chromosome combinations in gametes are formed by independent collections of chromosomes
- The process of fertilization in which one of the countless sperms fuses with an egg.
- A mutation in the gamete forming germ cells.
- Gene flow or the transfer of genes from one population of a species to another.

2. Define Evolution?

Evolution:

Evolution is the gradual alteration of a population's or species traits over time in response to the environmental conditions. Evolutionary changes can always be passed on to next generations.

3. Explain the theory of evolution by natural selection with examples.

Ans: Natural Selection is the process through which organisms who possess traits that are helpful for reproduction. Natural Selection theory is based on:

Over-production: All living creatures are very fertile so that their populations could grow quickly to produce large number of offspring.

Competition: Members of species' population will compete with one another for resources. Only



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the healthiest, fittest and strongest will survive and reproduce.

Variations: No two organisms are the same. They will not have the same opportunities to survive and reproduce as others.

Natural Selection: Only those organisms who live and reproduce can give rise to the next generation. And only those organisms with the desirable variations will live till reproductive age.

Observation Darwin made during his voyage on HMS Beagle:

One key observation Darwin made occurred while he was studying the specimens from the Galapagos Islands. He noticed the finches on the island were similar to the finches from the mainland, but each showed certain characteristics that helped them to gather food more easily in their specific habitat. He collected many specimens of the finches on the Galapagos Islands. These specimens and his notebooks provided Darwin with a record of his observations as he developed the theory of evolution through natural selection.

Analysing a Case study of Variation and Selection

The phrase "Changes in the environment can bring about changes in a population by natural Selection" is supported by following evidence.

Let's say a population of mice lives on a beach. In the sandy environment, a mouse with a white or beige fur color would blend in with the sand, making them less vulnerable to predators. Meanwhile, a dark brown mouse would stand out, making it extremely vulnerable to predation. These mice have a higher chance of dying before they get the chance to reproduce. The lighter mice will have a much greater chance of surviving and passing on their genes to the next generation.

4. Describe the Evidences of Evolution?

Ans: Evolution is the change in the physical characteristics of a species over numerous generations and it depends on the process of natural selection. The theory of evolution is based on the idea that all species are related and progressively change over time.

Fossils

They are the remains of the non-degraded parts of living beings found within rocks. The layer from which the fossil is recovered decides the age of the fossil. The fossils found deeper down are older compared to the ones found in the upper layers. They give an idea of the condition of the earth in that specific era. The study of fossils is called palaeontology. Thus, fossils are called paleontological evidences.

Comparative Anatomy

It explains that many organisms have the same ancestor and many different organisms evolved



as a result of natural selection or genetic drift. Two types of comparative anatomy were observed, namely homologous organs and analogous organs.

Homologous organs: These organs have the similar structure in different organisms but perform different functions. Thus, it clearly shows a common ancestry. This type of evolution is called divergent evolution. For example, the limbs of human, cheetah, whale, and wings of a bat. They all have different functions but have a similar structure indicating a common ancestor.

Analogous organs: These organs have different anatomy but perform similar function. This type of evolution is called convergent evolution. For example, The wings of birds have feathers for flight whereas, bats have wings made up of stretched skin and lack feathers. But, both serve the function of flight.

Evidence from Selective Breeding:

Selective breeding is when humans breed plants and animals for particular genetic characteristics so that they are different to their actual wild ancestors and are now beneficial to humans. In selective breeding characteristics are chosen by humans rather than the environment, so it is sometimes called artificial selection.

Main steps involved

Selective breeding takes place over many generations. The main steps for both plants and animals involve:

- Decide which characteristics are important enough to select.
- Choose parents that show these characteristics from a mixed population. They are bred together.
- Choose the offspring with the desired characteristics. Breed these together to produce the next generation.
- Repeat the process continuously over many generations, until all offspring show the desired characteristics.

Desired characteristics in plants:

- disease resistance in food crops
- wheat plants that produce lots of grain
- large or unusual flowers

Desired characteristics in animals:



- animals that produce lots of milk or meat
- chickens that lay large eggs
- domestic dogs that have a gentle nature

5. Differentiate b/w Natural Selection and Artificial Selection.

Natural Selection	Artificial Selection
Selection occurs when natural environment condition changes.	It is selected by man.
Varieties are produced by mutations.	Varieties are produced by selective breeding.
It operates in natural population.	It operates in domesticated population.
It is a slow process and takes thousand to millions of years.	It is fast process and results are immediate.
It is responsible for the great biological diversity.	It is responsible for new varieties different from starting generation.

6. Define Speciation.

Speciation:

The evolutionary process by which new biological species arise is called speciation. Two groups that originate from the same ancestor can become very different over time if they live in different places. When a species gets split into two geographical regions, the process of speciation starts.

